ELECTRICAL ITEMS

Qty	Type
276*	Receptacles, switches & lights
10*	Breakers (New Branch Circuit)
0	Conduit (Underground)
TBC	Equipment (Hardwired)

Panels

0	Service Panel 0 to 399 AMP
0	Service Panel 400 to 1,000 AMP
0	Service Panel over 1,000 AMP

Subpanel

4	Sub Panel 0 to 400 AMP

Qty	Motors
TBC	Motors < 3 HP/KW/KVA
TBC	Motors > 3-10 HP/KW/KVA
TBC	Motors > 10-50 HP/KW/KVA

EV Charger

0	EV Charger Residential (outlet, hardwire)

Temp Power

0	Temporary Service Pole
TBC	Temporary Distribution System
0	Generator

* Drawing takeoff. TBC = to be confirmed. See attached counting basis before submission.

MECHANICAL ITEMS

Qty	Type
	Furnace 8,000 BTU - Condenser 8 Ton
	Furnace 500,000 BTU - Condenser 40 Ton
	Furnace <500,000 BTU - Condenser >40 Ton
	FAU - Air Handler to 2,000 CFM
4	FAU - Air Handler to 10,000 CFM
	FAU - Air Handler over 500,000 CFM
	Evaporative Cooler (Swamp Cooler)
	Fireplace Appliance Vent
3	Mini Split Room Unit (Each)
3	Mini Split Outside Unit (Each)
	Cooling Coil

Qty	Type
4	Exhaust Fan (Restrooms, Laundry)
	Hood Vents (Range/Cooktop)
	Dry Duct
	Commercial Hood / Spray Booth
58	Registers
	Fire Damper
9	Duct System (Per Branch)

Misc

	Gas System (Alt more than 5')
	Electrical Disconnect (New/Relocate)

PLUMBING ITEMS

Qty	Fixtures
2	Water Closet (Toilet/Urinal)
	Bath Tub
	Shower
4	Lavatory (Restroom Sink)
	Sink (Kitchen, Commercial, Laundry, etc)
	Dishwasher
	Clothes Washer
1	Water Heater (Tankless add gas if upsizing)
	Repipe per Fixture

Gas

	Low Pressure (per outlet)
	Medium Pressure (per outlet)

Qty	Water Distribution
	Water Service - up to 1.5"
1	Water Service - 2" to 3"
	Water Service - over 3"
	Anti-Siphon Valve (Hose Bibb)
	Pressure Reulator

Sewer & Septic

	Sewer Repair (or Clean Out)
	Public Sewer Connection
	Building Sewer (ADU/Separate Structures)
	Septic Tanks / Seepage Pits
	Cesspool / Overflow Pits
	Grease Interceptor

Electrical quantity basis

BofA - Arrow Orange Hwy. | 903 W. Arrow Hwy., San Dimas, CA

Review draft: Filled electrical quantities use the October 5, 2026 electrical issue. Existing mechanical and plumbing entries are retained. Equipment/motor classifications and contractor temporary distribution are unresolved; this draft is not ready for permit submission.

1. Receptacles, switches & lights: 276 outlet/fixture locations

Component	Qty	Counting basis
Receptacle outlets	102	E01.00: 94 drawn symbols + 3 additional outlets for the (3)/(2) grouped quantities in NER. E03.01: 5 rooftop outlets. Each duplex/quad symbol counts as one outlet location.
Wall switches / dimmers / wall occupancy switches	20	E02.00: 14 keyed-note 2 controls, 5 keyed-note 5 controls, 1 keyed-note 6 switch. Ceiling sensors and room controllers excluded from this combined count.
Light fixtures, including exit signs	154	E02.00: R-14 = 49; R-15 = 14; R-19 = 27; R-11.4 = 8; R-11.8 = 6; R-21 = 2; D = 14; H = 11; L1 = 15; exit signs = 8. Duplicate H background labels are counted once.

2. Breakers, panels and zero entries

10 new breaker assemblies: E07.00 schedule note 3: A-32, A-33/35, A-34, A-36/38, A-37/39, A-40, A-42; B-30/32; F-1/3/5 and F-7/9/11. A multipole breaker counts once. Existing breakers marked note 2 are excluded, even where serving new loads. Confirm whether the form instead requires all new branch wiring on reused breakers.

4 subpanels: relocated A, B, C and D; existing 225A buses / 200A feeders. Existing F remains. No new service panel is shown. Underground conduit, residential EV charger, temporary service pole and generator: 0 shown in the issued permanent-work plans. Temporary distribution remains TBC: E00.00 general note 37 calls for contractor temporary wiring and lighting.

3. Equipment and motor entries: confirm category assignment

Connection inventory includes **4 RTUs, 3 outdoor heat pumps, 3 indoor fan-coil interconnections, 4 exhaust fans, 3 condensate pumps and 1 roll-down-door motor**. Other hardwired points include inverter, JACE, lighting contactor, thermafuser, mirrors, hand dryers, signage, flagscape, ATM surround, door hardware/actuators and water heater. Do not treat every J-box as hardwired equipment; many are security/data provisions.

The panel schedule gives provisional unit loads: HP-1/2 = 7.08 kVA each; HP-3 = 4.58 kVA; RTU-1 = 14.76 kVA, RTU-2 = 20.16 kVA, RTU-3/4 = 16.56 kVA each. These suggest 3 heat-pump units in the 3-10 range and 4 RTUs in the 10-50 range **if the City uses whole-unit scheduled kVA**. Verify actual nameplate ratings and classification; MCA and breaker amperes alone do not establish motor HP. M05.01 lists RTU indoor fans at 3 HP and exhaust fans at 29-45 W. Separate indoor-unit connections and integral motors/pumps may be treated differently.

The 4.5 kW water heater and powered signs also need classification. San Dimas publishes separate apparatus/appliance and sign fee categories (BE-22 through BE-29); confirm their mapping to this abbreviated form to avoid counting the same item twice. Mechanical disconnect quantity has not been changed.

4. Drawing issue to reconcile

E03.01 roof notes call for 55A/2P breakers for HP-1/2 and 35A/2P for HP-3; E07.00 schedules show 20A/2P for all three. This discrepancy does not change the three-unit count, but should be resolved before finalizing breaker requirements.

Sources: supplied Electrical.pdf (E01.00, E02.00, E03.00, E03.01, E07.00); matching Mechanical.pdf (M05.01).
City reference: sandimasca.gov published 2025-26 fee schedule, Electrical, BE-11 through BE-33. No fee calculation performed.